THERMODYNAMICS — ASSIGNMENT 01

37 Subjective Numericals · JEE Main PYQs · 4-page Print Edition

R = 8.3 J mol⁻¹ K⁻¹ | ln 2 = 0.693, ln 4 = 1.386, ln 8 = 2.079 | Work by gas is positive.

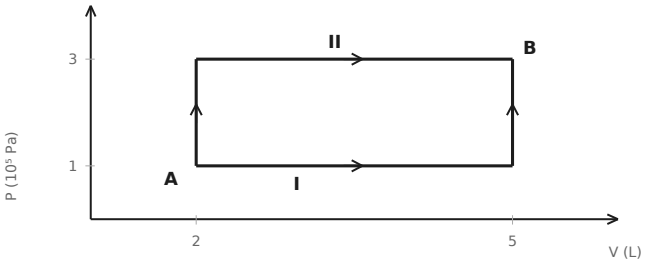
SECTION A — FIRST LAW & P–V WORK

1. A gas absorbs **1800 J** of heat and performs **650 J** of work. Find its change in internal energy and its final internal energy if the initial value is 2400 J.

2. **2 mol** of monatomic ideal gas are heated from **300 K to 450 K** at constant pressure. Calculate work done, increase in internal energy and heat supplied.

3. A monatomic ideal gas changes from **A (2 L, 1 × 10^5 Pa)** to **B (5 L, 2 × 10^5 Pa)** through an arbitrary path. Calculate the change in its internal energy.

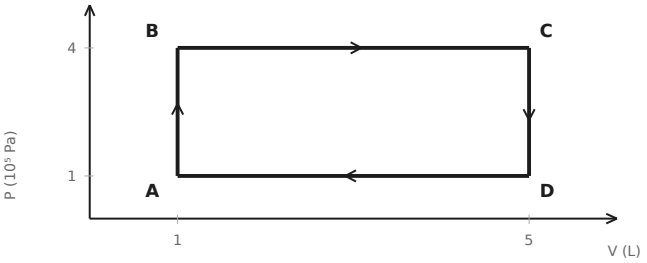
4. A monatomic ideal gas moves from **A to B** by two paths. Path I: constant-pressure expansion followed by constant-volume heating. Path II: constant-volume heating followed by constant-pressure expansion. Calculate W and Q for each path and $Q_{II} - Q_I$.



5. An ideal gas at **2 L and 4 × 10^5 Pa** expands isothermally to **8 L**. Determine its final pressure and work done by the gas.

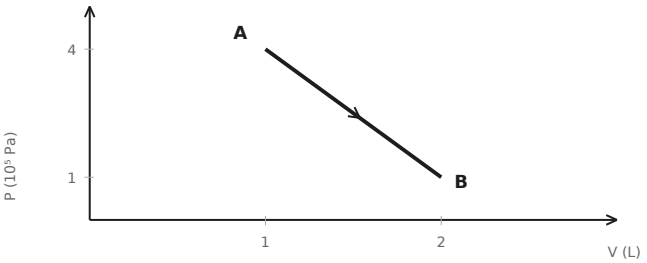
6. One mole of monatomic ideal gas at **300 K** is heated at fixed volume to **600 K**, then expands isothermally until its volume doubles. Find the total work and total heat absorbed.

7. An ideal gas executes the **clockwise rectangular cycle** A → B → C → D → A shown. Find net work done by the gas and net heat absorbed in one cycle.

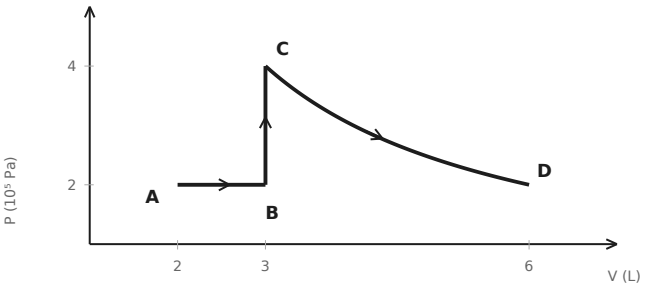


8. An ideal gas occupies **8 L at 1 × 10^5 Pa** and is slowly compressed isothermally until its pressure is eight times the initial value. Find final volume, work done by the gas and heat exchanged, with signs.

9. A monatomic ideal gas expands along the straight-line path A → B shown. Find work done, ΔU and heat exchanged. State whether the gas absorbs or releases heat.



10. A monatomic ideal gas follows A → B at constant pressure, B → C at constant volume, and C → D isothermally. Calculate the total work done, ΔU and heat absorbed.



SECTION B — HEAT CAPACITIES & GAMMA

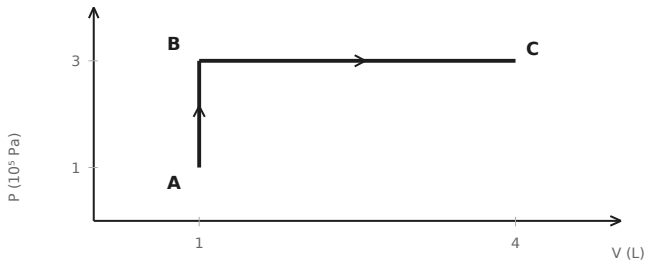
11. **2 mol** of diatomic ideal gas are heated from **293 K to 353 K**. Determine molar C_V and C_P, the heat required at constant volume and at constant pressure, and the difference.

12. **3 mol** of monatomic ideal gas expand at constant pressure, performing **1494 J** of work. Find ΔT, heat supplied and increase in internal energy.

13. **2 mol** of an unknown ideal gas absorb **3320 J** at constant pressure as the temperature rises from **200 K to 280 K**. Determine C_P, degrees of freedom f and γ.

14. An ideal gas with **γ = 4/3** absorbs **1600 J** at constant pressure. Find increase in internal energy, work done by the gas and number of active degrees of freedom.

15. A diatomic ideal gas undergoes A → B → C along the P–V path shown. Find the heat supplied during A → B, during B → C and over the complete process.



16. Monatomic gas X and diatomic gas Y separately follow the same straight-line P–V path from **A (2 L, 2 × 10⁵ Pa)** to **B (5 L, 3 × 10⁵ Pa)**. For each, calculate work and heat absorbed, and find $Q_Y - Q_X$.

17. An unknown ideal gas expands at constant pressure **3 × 10⁵ Pa** from **2 L to 5 L** and absorbs **2250 J**. Find W, ΔU, degrees of freedom f and γ.

18. A rigid **10 L** vessel holds a diatomic ideal gas at **1 × 10⁵ Pa**. Heat it until its pressure becomes **five times** the initial pressure. Find the heat supplied. (f = 5.)

19. **1 mol** of monatomic ideal gas follows **P ∝ V** from A (2 L, 1 × 10⁵ Pa) to B (4 L, 2 × 10⁵ Pa). Determine W, Q and the **molar heat capacity of this process**.

SECTION C — GAS MIXTURES

20. A mixture contains **2 mol helium** (monatomic) and **3 mol hydrogen** (diatomic). Find its effective degrees of freedom, molar C_V, C_P and γ_{mix}.

21. **3 mol** of monatomic gas are mixed with n mol of diatomic gas. The mixture has **γ = 29/19**. Determine n and effective degrees of freedom.

22. A rigid vessel holds **1 mol helium** and **2 mol nitrogen**, initially at **300 K**. The mixture is heated to **420 K**. Find Q, P_f/P_i and γ_{mix}.

23. A mixture of **2 mol helium** and **2 mol oxygen** is heated through **50 K** at constant pressure. Find heat supplied, work done, ΔU and γ_{mix}.

24. A rigid insulated vessel has two compartments: **1 mol helium at 600 K** and **2 mol nitrogen at 340 K**. The insulating partition is removed. Determine the final equilibrium temperature.

25. **2 mol** monatomic gas are mixed with n mol of non-linear polyatomic gas (f = 6). If $\boldsymbol{\gamma_{mix} = 7/5}$, find n and the molar C_V and C_P of the mixture.

26. A mixture of **3 mol monatomic** gas and **1 mol diatomic** gas moves A (1 L, 2 × 10⁵ Pa) → B (1 L, 4 × 10⁵ Pa) → C (3 L, 4 × 10⁵ Pa). Find γ_{mix} and heat supplied in each leg and in total.

27. **8 g helium** and **14 g nitrogen** are mixed and heated by **120 K** in a rigid vessel. Find the heat supplied and γ_{mix}. Molar masses: He = 4, N_2 = 28 g mol^{-1}.

28. A mixture of **2 mol helium**, **3 mol nitrogen** and **1 mol nonlinear polyatomic gas** (f = 6) is heated through **80 K at constant pressure**. Find the effective f, ΔU, W and Q.

29. In a **thermally insulated cylinder with a free piston**, 1 mol helium at 600 K and 2 mol nitrogen at 300 K are initially separated at the same constant pressure. The partition is removed and they mix at constant pressure. Find the final equilibrium temperature.

SECTION D — JEE PYQs (2022–2026) · JEE MAIN ONLY

30. A gas with **8 active degrees of freedom** performs **150 J** of work during expansion at constant pressure. Calculate the heat supplied.

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31. A system receives heat at **1000 J s^{-1}** and does work at **200 J s^{-1}**. At what rate does its internal energy increase?

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32. A diatomic ideal gas (rotational modes active, vibration neglected) absorbs **735 J** during constant-pressure expansion. Find the increase in internal energy.

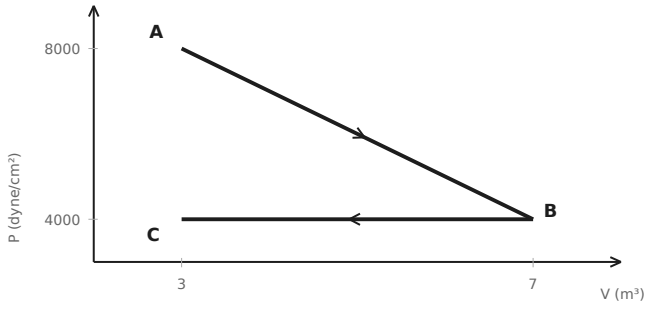
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33. **48 J** of heat are supplied to **1 mol helium**, raising its temperature by **2 K**. Determine the work done by the gas.

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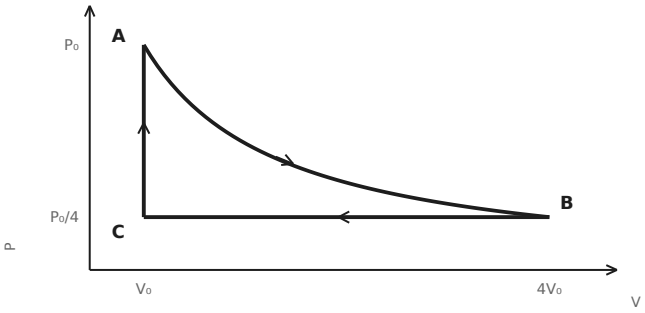
34. A gas moves A → B along the straight P–V line shown, then B → C at constant pressure. Calculate net work in joules. Use **1 dyne cm^{-2} = 0.1 Pa**.

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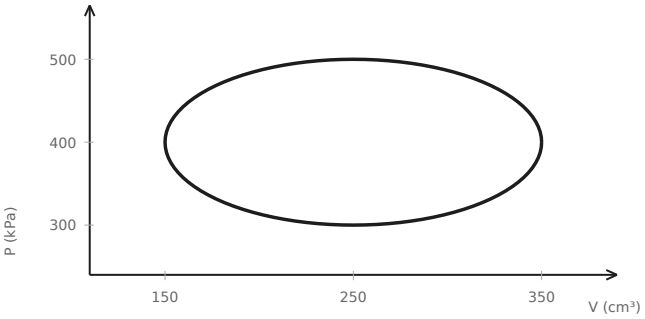
35. Starting at (P_0,V_0), an ideal gas expands **isothermally to 4V_0**, compresses isobarically to V_0, and returns to the initial state by isochoric heating. Find net heat absorbed in one cycle in terms of P_0V_0.

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36. A gas follows the closed elliptical P–V loop shown. Determine the **magnitude** of net work (J) and the numerical value of W/(10^{-1} J). Take π = 3.14.

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37. **5 mol** of an ideal gas are heated at constant volume from **10°C to 20°C**. Given C_P = 8 cal mol^{-1} K^{-1} and R = 8.36 J mol^{-1} K^{-1}, find ΔU in calories. (1 cal = 4.18 J.)

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ANSWER KEY

Final answers only · No solutions or explanations

SECTION A — FIRST LAW & P–V WORK

1. $\Delta U = +1150$ J; $U_f = 3550$ J

2. $W = 2490$ J; $\Delta U = 3735$ J; $Q = 6225$ J

3. $\Delta U = +1200$ J

4. I: $W = 300$ J, $Q = 2250$ J; II: $W = 900$ J, $Q = 2850$ J; difference = 600 J

5. $P_f = 1 \times 10^5$ Pa; $W = 1108.8$ J

6. $W = 3451.14$ J; $Q = 7186.14$ J

7. $W_{net} = Q_{net} = +1200$ J

8. $V_f = 1$ L; $W = Q = -1663.2$ J

9. $W = +250$ J; $\Delta U = -300$ J; $Q = -50$ J (released)

10. $W = 1031.6$ J; $\Delta U = 1200$ J; $Q = 2231.6$ J

SECTION B — HEAT CAPACITIES & GAMMA

11. $C_V = 20.75$; $C_P = 29.05$ J mol^{-1} K^{-1}; $Q_V = 2490$ J; $Q_P = 3486$ J; difference = 996 J

12. $\Delta T = 60$ K; $Q = 3735$ J; $\Delta U = 2241$ J

13. $C_P = 20.75$ J mol^{-1} K^{-1}; $f = 3$; $\gamma = 5/3$

14. $\Delta U = 1200$ J; $W = 400$ J; $f = 6$

15. $Q_{AB} = 500$ J; $Q_{BC} = 3150$ J; $Q = 3650$ J

16. Both: $W = 750$ J; $Q_X = 2400$ J; $Q_Y = 3500$ J; difference = 1100 J

17. $W = 900$ J; $\Delta U = 1350$ J; $f = 3$; $\gamma = 5/3$

18. $Q = 10{,}000$ J

19. $W = 300$ J; $Q = 1200$ J; $C = 2R = 16.6$ J mol^{-1} K^{-1}

SECTION C — GAS MIXTURES

20. $f = 21/5$; $C_V = 17.43$; $C_P = 25.73$ J mol^{-1} K^{-1}; $\gamma = 31/21$

21. $n = 2$ mol; $f = 19/5$

22. $Q = 6474$ J; $P_f/P_i = 1.4$; $\gamma = 19/13$

23. $Q = 4980$ J; $W = 1660$ J; $\Delta U = 3320$ J; $\gamma = 3/2$

24. $T_f = 400$ K

25. $n = 4$ mol; $C_V = 20.75$; $C_P = 29.05$ J mol^{-1} K^{-1}

26. $\gamma = 11/7$; $Q_{AB} = 350$ J; $Q_{BC} = 2200$ J; $Q = 2550$ J

27. $Q = 4233$ J; $\gamma = 27/17$

28. $f = 9/2$; $\Delta U = 8964$ J; $W = 3984$ J; $Q = 12{,}948$ J

29. $T_f = 7200/19$ K ≈ 379 K

SECTION D — JEE PYQs (2022–2026) · JEE MAIN ONLY

30. $Q = 750$ J

31. $dU/dt = 800$ W

32. $\Delta U = 525$ J

33. $W = 23.1$ J

34. $W = 800$ J

35. $Q_{net} = P_0V_0(\ln 4 - 3/4) = 0.636\ P_0V_0$

36. $|W| = 31.4$ J; W/(0.1 J) by magnitude = 314

37. $\Delta U = 300$ cal